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ORIGINAL RESEARCH

Automatic extraction and collection of legal principles from CJEU decisions on Fiscal State Aid

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Abstract
Legal principles stated in judicial decisions are a key reference for judges and legal professionals. However, typically, decisions are long and complex documents. The automatic extraction of legal principles therein may thus be beneficial to many. Unfortunately, this is an under-investigated task for which few resources and implementations exist, especially with regard to EU law. To promote research in this domain, we develop a novel corpus of 64 judicial decisions on Fiscal State Aid by the CJEU (the Court of Justice of the European Union), in which legal principles are identified and tagged. With this new resource, we conduct extensive experimentation to identify the best combinations of representation methods, classifiers and data augmentation methods for the task at hand, obtaining an F1 score of 0.72. Additionally, we consider the collections of legal principles obtained by running the extractive task and collating the output paragraphs together. We show, through the use of lexical metrics and human evaluation, that the quality of generated collections is comparable to that of expert-made ones.

Keywords Legal principles · Text classification · Data augmentation

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1 Introduction

Judicial decisions are a key source of information for lawyers, judges and other legal practitioners. *Legal principles* are among the most important elements of such decisions. These are criteria stated (or referred to) by the judges, except for literal citations of legislative or regulatory sources. An example of legal principle is:

> *As regards appraisal of the condition of selectivity, it is clear from settled case-law that Article 87(1) EC requires assessment of whether, under a particular legal regime, a national measure is such as to favour 'certain undertakings or the production of certain goods' in comparison with others which, in the light of the objective pursued by that regime, are in a comparable factual and legal situation.*[1]

Legal principles are important to judges, who must justify their decisions by stating the relevant legal principles, often by referring to earlier cases where they were used or introduced. Legal practitioners also use legal principles when arguing to support any legal claim. For instance, a proposed decision in a new case may be justified by arguing that it follows from the criteria used in precedents, or can be disqualified by holding that it is in conflict with them. Such principles are relevant not only in common law systems, where precedents are binding, but also in civil law systems, where past judicial decisions, especially by the highest courts, have a "persuasive authority".

The specification of legal principles is a key aspect of the case law of the Court of Justice of the European Union (CJEU), which is tasked with interpreting and applying EU law. In this context, the notion of "legal principle" includes both *rationes decidendi*, the rulings that decisively contribute to the decision, and *obiter dicta*, legal criteria mentioned incidentally or in dissenting opinions. The CJEU formulates principles that specify, for example, what facts fall under an EU-law concept or determine the meaning of legislative rules, possibly restricting or extending their scope.

The automated extraction of these principles would be important from a theoretical perspective, to support a broader legal understanding of case law domains, enabling legal scholars and professionals to quickly grasp the innovative content of judicial decisions, i.e., the rulings, that define the state of the law and shape its evolution. It would also have practical benefits by supporting a work-intensive and time-consuming task. Moreover, automatically extracted collections of legal principles may: (a) help understanding the key content of judicial decisions, (b) facilitate the summarization of judicial decisions, and (c) enable the automated retrieval of precedents based on the same or related principles. All this could facilitate the cross-case application of legal reasoning, promote doctrinal and judicial consistency and thus mitigate the risk of conflicting interpretations.

In this work, we address the automated identification and extraction of general and domain-specific legal principles from the CJEU decisions on Fiscal State Aid. While

[1] Joint Cases of European Commission (C-106/09 P) and Kingdom of Spain (C-107/09 P) v Government of Gibraltar and United Kingdom of Great Britain and Northern Ireland.

the US legal system has a tradition of producing hand-notes to judicial opinions, which also contain legal principles, by contrast, little work has been done on identifying legal principles in the EU law. It is thus an under-investigated area. Moreover, the discipline has been entirely developed through the CJEU judgments (Schön 2016).

Our contribution consists of:

- a new set of guidelines for identifying general and domain-specific legal principles in judicial decisions;
- a novel corpus consisting of 64 CJEU judicial decisions on Fiscal State Aid, where legal principles have been manually identified;
- an experimental study regarding the automatic identification and extraction of principles; our study includes 11 baselines, 3 Large Language Models (LLMs), and 72 combinations of representations, data augmentation techniques, and classifiers, totaling 86 models;
- an evaluation of the quality of the extracted principles, based on the ROUGE lexical metrics (Lin 2004);
- an in-depth human evaluation of (i) the quality of the extract principles and (ii) the coherence of the collections of principles, relatively to the analysed judicial decisions.

From the Natural Language Processing (NLP) perspective, the task is challenging since recognizing legal principles implies the ability to understand the legal content of a portion of text. To reach the best results, machine learning-based methods typically require plenty of data and balanced datasets (Russell and Norvig 2010). This is particularly difficult in the legal domain, where datasets are typically small and unbalanced. However, also the creation of novel resources includes several challenges from a legal-informatics perspective. The first one amounts to specifying what should count as a legal principle, given that principles are often context-dependent. Second, creating a novel annotation framework requires a substantial understanding of the law and argument analysis. Finally, the annotation process requires considerable effort because CJEU decisions are long and complex documents; moreover, it must be performed by experts.

The paper is structured as follows. In Section 2, we provide a short NLP background and describe the relevant state of the art. In Section 3, we introduce the particulars of the State aid law/discipline and the role of CJEU judgments in the fiscal State aid domain. Section 4 presents the corpus and the annotation procedure. Section 5 explains the experimental methodology. Results are discussed in Section 6. In Section 7, we present our results on the evaluation of our method for the creation of collections of legal principles. We include intrinsic evaluation based on standard metrics and a human study. Section 8 is dedicated to considerations concerning limitations, impact, and ethics of our work. Section 9 concludes.

2 Background and related work

2.1 NLP background

Historically, NLP has been addressed by traditional machine learning methods like logistic regression, decision trees, support vector machines, and shallow neural networks, combined with approximate text representations (e.g., bag-of-words) (Goldberg 2018). While these methods bring relevant advantages like greater transparency and small training costs, they rely on text representations that do not enable a rich semantic representation.

Deep encoder models like transformers aim at narrowing this gap by capturing contextual dependencies of word meaning and defining text representations that adapt well to different tasks without further adjustments (Vaswani et al. 2017), and that can be combined with traditional machine learning methods. However, adapting deep encoders to specific domains requires costly ad-hoc training.

More recently, large generative models have been preferred to deep encoders due to their notable adaptability to different tasks and domains without incurring additional training costs or the need for new data. Large generative models view each task as a text generation problem where the textual input contains both domain-specific data and task instructions to guide the model. This mechanism, denoted as test-time compute or prompting (Liu et al. 2023), allows the model to adapt to new tasks by changing task instructions only in the input, resulting in a lower need for data (Hu et al. 2022). Despite their widespread application, large generative models may still underperform compared to deep encoder models in text classification tasks (Zhou et al. 2024). For this reason, both methods are often evaluated on the same data to provide a broad empirical evaluation of NLP systems.

Readers interested in more details about NLP and its applications may refer to Jurafsky and Martin (2025).

2.2 Legal principle detection

The text classification task we address in this paper is the identification and extraction of legal principles from judicial decisions, specifically from the CJEU in the Fiscal State Aid domain. This task is related to information retrieval (IR) (Sansone and Sperlí 2022), since it involves finding the relevant sentences in the text, to argument mining (AM) (Mochales and Moens 2011; Lippi and Torroni 2016), since legal principles work as warrants in legal arguments, as well as to the identification of legal factors (Gray et al. 2022, 2024; Mumford et al. 2022). The methods developed in these research fields may also be useful to address the detection of legal principles. However, we will focus on studies that specifically target legal principles.

Relevant works include research on identifying authoritative rules of decisions in common law (*ratio decidendi*) and distinguishing these from other legal criteria mentioned in judicial decisions (*obiter dicta*). Branting (1993) uses argument graphs (i.e., reduction-graphs) to model *rationes decidendi* in the US tort cases. With the same goal, Plug (2000) focuses on linguistic clues (hedging expressions such as '*anyway*', '*if only because*') for identifying non-authoritative warrants (*obiter dicta*) in Dutch,

English and American decisions. In the Information Retrieval context, Grabmair et al. (2015) provide a framework for the automatic annotation of judicial sentences as pertaining to legal principle or to evidence about facts. The distinction relies on features and entity types, identified through manually coded UIMA rules. The sentences annotated as legal principles receive preferential treatment when assessing the relevance of document to a search query. Additionally, some recent work (Achena et al. 2023; Preti et al. 2024) focuses on extracting legal principles in Italian case law through summarization techniques.

More recently, the detection of legal principles in cases has been performed using NLP and machine learning (ML) methods. Shulayeva et al. (2017) annotated 50 common law reports by distinguishing between factual statements and legal principles. The classification of a sentence is based on a combination of linguistic features (e.g., part of speech tags, unigrams, etc.) as well as on contextual clues, e.g., whether the sentence has been taken from earlier cases (*citing sentences*), which favour considering it as a principle. No further distinction between binding and non-binding principles was made. Automated classification using Naive Bayesian Multinomial achieved 82% precision on both facts and principles. Error analysis revealed two main challenges with the task: (i) it is linguistically challenging to distinguish between cited principles and discussions of principles by judges in a new case; (ii) some sentences can contain both facts and principles making it difficult to assign a single label.

In contrast, Valvoda and Ray (2017) focused on *cited paragraphs* (passages cited in later cases) and distinguished between binding and non-binding legal principles. They used a well-cited case[2] and cases citing it. To distinguish between *rationes decidendi* and *obiter dicta*, human experts applied the Wambaugh's Inversion Test (Branting 1994): a counterfactual test for assessing whether a removal of the principle affects the decision in the case. An automated approach combining the identification of principles and cited paragraphs was deployed. The approach yielded 72% accuracy for the identification of *rationes decidenci*. The single task of legal principle classification (both rationes and dicta) yielded a 96% accuracy. A downside of this approach is that it only works with sufficiently well-cited cases.

Despite their theoretical and legal importance, no empirical work has been conducted in the context of the European Court of Justice. To the best of our knowledge, we are the first to propose an experimental study on the automatic extraction and collection of legal principles in this domain.

2.3 EU law datasets

There is little work regarding automatic extractions of legal principles in EU law. Existing benchmarks (Chalkidis et al. 2022) and automated analysis are mostly focused on the decisions of the European Court of Human Rights including studies on case outcome predictions (Poudyal et al. 2020; Medvedeva et al. 2020; Chalkidis et al. 2020), citation networks (Šadl and Olsen 2017; Lupu and Voeten 2012), and argument mining (Habernal et al. 2023; Poudyal et al. 2020).

[2] *Stack v Dowden* [2007] UKHL 17.

However, there is a recent growing interest in the principles used by judges in CJEU decisions, probably due to the increasing reliance on past interpretations in the CJEU. This is certainly the case of the State Aid domain, where CJEU case law is paramount. In this context, our work aims to fill a research gap, and contribute to supporting the CJEU and national judges. All CJEU decisions are published on the EUR-LEX platform, but little has been done in terms of developing specific corpora for NLP analysis. One exception is the Demosthenes corpus, (Grundler et al. 2022; Santin et al. 2023) which contains 40 decisions on Fiscal State Aid by the CJEU (covering the period 2000-2018). These decisions have been manually annotated for the purpose of argument structure prediction. Part of these documents are included in our dataset, but we address a different task providing a new and independent annotation layer.

2.4 Data augmentation

The performance of machine learning-based methods is notoriously dependent on the amount of data on which they are trained. Moreover, the diversity of the data plays an important role in the ability of the models to generalize enough to be deployed in real-world situations. In the legal domain, the process of collecting or creating new data is long and expensive. A possible approach to increasing the number of training data, enhancing their diversity, and addressing data scarcity and class imbalances is data augmentation. It consists of manipulating the samples within the dataset to create new synthetic training data.

Data augmentation is applied in various fields including computer vision, where the manipulation is more straightforward, while the discrete nature of the text space makes the generation of the new samples more complicated (Feng et al. 2021). Textual data augmentation is an ongoing area of research (Shorten et al. 2021) with work including rule-based approaches such as word-level token insertion, deletion, swapping or syntactic manipulations (Wei and Zou 2019; Jin et al. 2020; Min et al. 2020), embedding-based augmentation (Wang and Yang 2015) and feature space augmentation (Cheung and Yeung 2021). Other works instead have used pre-trained transformer-based models using the name of the classes to condition the generation (Kumar et al. 2021; Wu et al. 2019).

Lastly, recent work on augmentation features examples where generative models are utilized due to their success in neural text generation tasks (Abonizio and Junior 2020; Anaby-Tavor et al. 2020; Gupta 2019).

Several of these techniques have been adopted in works concerning the legal domain. Yan et al. (2019) used random scrambling, deletion and insertion of sentences to generate new samples, Guo et al. (2020) employed a pipeline of stop-word removal, backtranslation, and replacement with synonyms to generate test data. For what concerns language models, Peric et al. (2020) explored using auto-regressive transformer models to generate judicial opinions, Ghosh et al. (2023) used an encoder-decoder language model pre-trained on a text denoising objective to augment the legal documents, and Sheik et al. (2024) explored GPT models for generating new samples via few-shot prompting.

In our work, we rely on the approach proposed by Perçin et al. (2022). In their paraphrasing-based methods, they experiment with several techniques to replace the words of a sentence with a semantically related alternative. Their results indicate that the best approach is to combine symbolic knowledge from resources such as WordNet with information from dense embedding similarity. The former is used to gather a set of plausible replacement words, the latter to select the most semantically related one.

3 Fiscal State Aid

In selecting the legal domain, we chose to focus on CJEU decisions on Fiscal State Aid, since (i) the field is relatively new and under-investigated; (ii) the discipline has been entirely developed through the CJEU judgments, providing interpretations that strongly influence the application of fiscal State aid law by Member State; (iii) the decisions contain a diverse set of legal principles; and (iv) have a standard (although not fixed) structure, which facilitates the principles extraction.

The concept of State Aid was introduced by the EU Treaty (the former EEC Treaty) in 1957, and refers to any form of influence by the State on the competition in the internal market. Its purpose is to constrain the Member States from granting undue benefits to certain economic operators. State aid is – as a general rule – prohibited, unless it is justified by reasons of general economic interests (see Article 107(2) and (3) TFEU).

More specifically, under article 107(2) of the Treaty, State aid is allowed when: (i) it fulfills objectives of common interest (services of general economic interest, social and regional cohesion, employment, research and development, sustainable development, promotion of cultural diversity, etc.); or (ii) it represents the appropriate instrument to correct certain "market failures". For instance, dealing with a market failure may, in some cases, offsets the distorting effects of competition: in such cases the aid is considered as compatible.

Under the general definition provided by article 107(1) TFEU, a State aid consists in an advantage conferred by national public authorities "through State resources, in any form whatsoever, which distorts or threatens to distort competition, by favouring certain undertakings or the production of certain goods". Thus, a measure can be considered as a State aid only if all the following elements are identified: (i) a support is granted by the State or through State resources; (ii) it favours one or more undertakings, i.e., there is a selective advantage; (iii) it (has the potential to) distort competition; and (iv) it affects trade between EU countries.

Fiscal State Aids concern national fiscal measures. As clarified by the CJEU, the concept of fiscal aid is wider than that of subsidies since it embraces not only a positive benefit (e.g., subsidies) but also intervention which, in any forms, mitigate charges normally included in the budget of an undertaking and which, without being subsidies in the strict sense, are similar in character and have the same effect (see Case C-387/92 Banco Exterior (para 13)). This domain heavily relies on CJEU case-law interpretation, making it a good candidate for automated support in extracting the relevant legal principles and their interpretation.

4 Corpus

We present a novel corpus of Fiscal State Aid decisions by the CJEU with manually identified legal principles. It contains 64 decisions in English, taken from the EUR-LEX database.[3] Our dataset includes all the decisions adopted by the CJEU between 1992 (the year when the CJEU inception as the Court of Appeal in this domain) and 2022. Most of them (55 out of 64) are appeal decisions (for State Aid decisions), while the others are relevant preliminary rulings (focused on interpretation of EU law) and actions for annulment. Table 1 reports on corpus statistics.

4.1 Document structure

CJEU decisions present a standard, even though not fixed, structure. They usually include the following six sections:

1. *The Preamble*, containing information on the parties – i.e., the Commission and a member State and/or a private competitor– the appealed judgement of the Court of First Instance, and the composition of the Court.
2. *The Case background*, including facts and the procedural case history before the General Court.
3. *The judgement under appeal*, reporting the assessment of the General Court in the first instance decision.
4. *The Grounds of Appeal*, typically divided in:

    1. the *Arguments of the Parties*, supporting or attacking each ground of appeal; and
    2. the *Findings of the Court*, i.e. the Court reasoning process, characterised by a set of argument chains, which lead to conclusions with regard to each parties' claim.

3. *Costs*, i.e., the attribution of costs.
4. The *Ruling*, i.e., the final decision and orders to the parties.

For the purpose of legal principle extraction, the most relevant section is the "Findings of the Court", since it contains interpretations of the rules and discussions about past cases.

4.2 Annotation method and guidelines

Legal theory presents different conceptualizations of legal principles and approaches to using them in legal reasoning (Goodhart 1959; Raz 1972; Dworkin 1977; Schauer 1991; Alexy 2000). However, this distinction is beyond the scope of this paper, which focuses on identifying all legal criteria stated by the CJEU to reach and justify its decisions. We use the term "legal principle" to refer to all such criteria. Thus, this

[3] https://eur-lex.europa.eu/collection/eu-law/eu-case-law.html

Table 1 Dataset composition. The length is measured as the number of characters

Splits	Documents	# Samples		Avg. Samples per Doc.		Avg. Length of Samples	
		NEG	POS	NEG	POS	NEG	POS
Training	38	5,460	398	143.7	10.5	339±290	654±244
Validation	13	1,712	103	131.7	7.9	318±269	556±210
Test	13	1,805	113	138.8	8.7	318±286	640±236
Total	64	8,977	614	140.3	9.6	331±286	635±239

notion includes all criteria pertaining the interpretations of legislation and case-law, only excluding excluded literal citations of legislative sources. Note that we include in legal principles both criteria stretching over different domains of the law, and domain-specific ones, pertaining to tax law and more precisely the Fiscal State Aid discipline. Also note, that, since in EU law there is no doctrine of binding precedent and CJEU decisions do not have dissenting opinions, we do not need to relate to the distinction between *ratio decidendi* and *obiter dicta* (MacCormick 2005; Lamond 2005; Duxbury 2008).

In CJEU decisions, each principle is usually contained in a single paragraph, where it can stretch over multiple sentences. Thus, in identifying legal principles, each paragraph could be analysed individually, without the need of focusing on the context. For the identification of legal principles we adopted a practical perspective, based on the analysis of (a) linguistic indicators, including keywords and word patterns; (b) context indicators; and (c) legal contents. Typically, when more than one indicator is present in a sentence, it is extremely likely that the sentence contains a legal principle.

The annotation process was carried out by two legal experts who hold a PhD in Law and were selected based on their expertise in this legal domain. The Inter-Annotator Agreement was measured on a sample subset of 124 principles independently annotated by both experts, obtaining a result of Cohen's $\kappa = 0.87$, which represents almost perfect agreement (Cohen 1960).

4.2.1 Linguistic indicators

Legal principles are often characterised by deontic modality expressions, such as *must* for obligation, *must not* for prohibition, or *may* for permission, which contrast with epistemic modalities for necessity and possibility. Consider the following examples:

> *[...]The concept of State aid must be applied to an objective situation, which must be appraised on the date on which the Commission takes its decision. (Case C-341/06 P, para. 9)*

> *Benefits may include not only positive benefits such as subsidies, loans or direct investment in the capital of undertakings, but also interventions which, in various forms, mitigate the charges which are normally included in the budget of an undertaking [...] (C-341/06 P, para. 123)*

The set of keywords and word patterns signaling or introducing a legal principle includes, for instance, “in principle”, “within the meaning of”, “the intention of”, “it must be noted that”, “it must be recalled that”, “it must be borne in mind that”. Consider the following examples:

The intention underlying Article 87(1) EC and the private investor test is thus to prevent the recipient public undertaking from being placed, by means of State resources, in a more favourable position than that of its competitors. (Case C-124/10 P, para. 90)

As regards that condition, that the advantage must be granted directly or indirectly through State resources, it must be recalled that measures not involving a transfer of State resources may fall within the concept of “aid”, within the meaning of Article 107(1) TFEU. (C-656/15 P, para. 43)

[...] in principle, aid intended to release an undertaking from costs which it would normally have to bear in its day-to-day management or normal activities distorts the conditions of competition. (Case C-494/06 P para. 54)

It should be noted that deontic modality expressions are often combined with keywords and word patterns, as shown in the following example:

[...] the concept of a “regulatory act which ... does not entail implementing measures”, within the meaning of the final limb of the fourth paragraph of Article 263 TFEU, is to be interpreted in the light of that provision’s objective, which, as is clear from its origin, consists in preventing an individual from being obliged to infringe the law in order to have access to a court.[...] (Case C-274/12, para. 27)

4.2.2 Context indicators

Context indicators may be used to signal or exclude the presence of a legal principle. In particular, citations from existing case law and references to legal precedents are often used by the CJEU to recall settled principles, so as to reinforce its arguments supporting the decision. In particular, if a sentence is followed by a reference to a past case, it will likely convey a principle from that case. Consider the following example:

It must also be taken into consideration that the particularly strict nature of the State aid regime under the ECSC Treaty sets it apart from the State aid regime under the EC Treaty. (C-408/04 P, para.105)

Conversely, the occurrence of linguistic indicators such as “according to the appellant”, or “according to the Commission” usually suggest that the Court merely recalls the arguments of the parties, indicating that the following sentence is not a decision criteria endorsed by the Court. As an example, consider the following statement:

According to the appellant, the particular nature of the review of State aid means that, when the court carrying out the review is called on to assess the facts of the case in the light of legal criteria, it must limit itself to satisfying itself that there is no manifest error of assessment. (Case C-83/98 P, para. 24)

4.2.3 Legal content

Linguistic and context indicators alone may not suffice in detecting legal principles. We identify several criteria based on the legal content of the analysed cases to further guide the detection task. In this regard we distinguish between (a) general criteria, which can be applied to any legal area, and (b) domain-specific criteria, which refers to concepts strictly pertaining the Fiscal State Aid domain.

General criteria Legal principles often specify circumstances and conditions either extending or limiting the application of legal concepts and rules. Consider the following example:

[...] the mere fact that a measure may exercise an influence on the competitive relationships existing on the relevant market and that the undertaking concerned was in a competitive relationship with the addressee of that measure cannot in any event suffice for that undertaking to be regarded as being individually concerned by that measure. (Case C487/06 P, para. 47)

On the other hand, statements merely mentioning the content of legislative provisions are not considered principles, since they do not express any kind of legal interpretation. Consider the following example:

Moreover, Article 15 of Regulation No 659/1999 provides that recovery of illegal aid is subject to a limitation period of 10 years, beginning on the day it is granted. (Case C-408/04 P, para. 102)

Domain-Specific criteria It is possible to distinguish between principles related to procedural and substantive matters. In both cases, adopting a domain-specific perspective facilitates the identification of legal principles. Procedural matters may concern whether the party (i) is individually concerned in the case, i.e., whether the applicant-competitor is one of the undertakings being affected by a State aid decision and (ii) has an interest in the procedure, i.e., the capability of the action to bring an advantage to the appellant. Procedural matters also cover certain general residual issues, such as, whether the appealed Court was competent in deciding the case or whether the types of complaint raised by the parties are admissible. The following examples illustrate the three said categories of procedural issues:

[...] persons other than those to whom a decision is addressed may only claim to be individually concerned if that decision affects them by reason of certain attributes which are peculiar to them or by reason of circumstances in which they are differentiated from all other persons and by virtue of these factors dis-

tinguishes them individually just as in the case of the person addressed. (Case C-298/00 P, para. 36)

Where a regulatory act directly affects the legal situation of a natural or legal person without requiring implementing measures, that person could be denied effective judicial protection if he did not have a legal remedy before the EU judicature for the purpose of challenging the lawfulness of the regulatory act. In the absence of implementing measures, a natural or legal person, although directly concerned by the act in question, would be able to obtain judicial review of the act only after having infringed its provisions, by pleading that those provisions are unlawful in proceedings initiated against them before the national court. (Joined Cases C-622/16 P and C-624/16 P, para. 58)

[...] the obligation to state reasons does not require the General Court to provide an account which follows exhaustively and one-by-one all the arguments put forward by the parties to the case, the General Court's reasoning may therefore be implicit on condition that it enables the persons concerned to know why it has not upheld their arguments and provides the Court of Justice with sufficient material for it to exercise its power of review. (Case C-100/16, para. 32)

Legal principles related to substantive matters may concern (i) the interpretation of the four criteria under which a fiscal measure should be considered a(n) (unlawful) State aid – i.e., the public origin of the supportive measure, the presence of an economic advantage, its selectivity, and its (potential) distorting effect on competition– (see section 3); (ii) whether a change to an approved aid measure affects its compatibility with the internal market, thus turning into a *new aid* that has to be re-notified to the European Commission for authorization; or (iii) general substantial issues related to the general State aid discipline. The latter is a residual category addressing, for instance, the statement of reasons required by the Commission in order to qualify a measure as state aid; general requirement for the classification of aids; and settlement on the recovery. The following examples respectively illustrate legal statements interpreting the aforementioned criteria:

The applicability of the private investor test ultimately depends, therefore, on the Member State concerned having conferred, in its capacity as shareholder and not in its capacity as public authority, an economic advantage on an undertaking belonging to it. It follows that, if a Member State relies on that test during the administrative procedure, it must, where there is doubt, establish unequivocally and on the basis of objective and verifiable evidence that the measure implemented falls to be ascribed to the State acting as shareholder. (Case C-124/10 P, para. 81, 82)

[...] the selectivity requirement differs depending on whether the measure in question is envisaged as a general scheme of aid or as individual aid. In the latter case, the identification of the economic advantage is, in principle, sufficient to support the presumption that it is selective. By contrast, when examin-

ing a general scheme of aid, it is necessary to identify whether the measure in question, notwithstanding the finding that it confers an advantage of general application, does so to the exclusive benefit of certain undertakings or certain sectors of activity. (Case C-15/14 P, para. 59)

[...] the only situation recognised by the Court's case-law in which the finding that an economic advantage has been granted does not lead to the measure at issue being categorised as State aid within the meaning of Article 107(1) TFEU is that in which a State measure represents the compensation for the services provided by undertakings entrusted with performing a service in the general public interest in order to discharge public service obligations, in accordance with the criteria established in the judgment of 24 July 2003, Altmark Trans and Regierungspräsidium Magdeburg (C-280/00, EU:C:2003:415).(Case C-211/15 P, para. 44)

[...] aid intended to release an undertaking from costs which it would normally have to bear in its day-to-day management or normal activities distorts the conditions of competition. (Case C-128/16 P, para. 84)

As an example of a legal principle concerning the concept of *new aid*, consider the following case:

[...] the Commission must, where appropriate, take into account any cumulative effect of the earlier unlawful aid that has not been repaid and the new aid and, second, it can find that the new aid is compatible with the common market only when the evidence at its disposal enables it to reach such a conclusion. (Case C-480/09 P, para 97)

In the following we report an example of a principle addressing a general substantial issue within the State aid discipline:

[...] the definition of aid is more general than that of a subsidy. It includes not only positive benefits but also measures which, in various forms, mitigate the charges which are normally included in the budget of an undertaking and which thus, without being subsidies in the strict sense of the word, are similar in character and have the same effect. (Case C-81/10 P, para. 16)

5 Method

We design the automatic extraction of legal principles as a paragraph-level binary classification task. The aim is to discriminate between paragraphs that contain legal principles (positive samples) and those that do not (negative samples). We consider each paragraph individually, without any context information. This choice is motivated by the length of the documents, the limited size of the input of neural models, and the observation that legal principles are contained in a single paragraph.

We consider three possible methods of classification: a set of baselines, machine learning-based classifiers trained for the task, and LLMs used in zero-shot setting. For training the classifiers, we follow three steps, summarized in Fig. 1. First, we generate 8 training sets through oversampling and data augmentation. Then, we encode each paragraph using Transformer-based encoders, obtaining 24 encoded training set. Finally, we train 3 types of classifiers over each encoded training set, obtaining 72 trained models.

We split the corpus into training, validation, and test sets with ratios of approximately 60%, 20% and 20%, resulting respectively in 38, 13, and 13 documents.

5.1 Input representation

We pre-process the text to remove tags and new line characters, then we tokenize it. We experiment with 3 different popular state-of-the-art encoders in the legal domain: BERT,[4] LEGAL-BERT,[5] and Sentence-BERT.[6] BERT (Devlin et al. 2019) is a widely adopted general-purpose pre-trained large language model based on Transformers (Vaswani et al. 2017) and attention (Galassi et al. 2021). We follow standard practice and use the embedding of the <CLS> token for sentence representation. We tokenize texts using 512 maximum token length. LEGAL-BERT (Chalkidis et al. 2020) is a BERT-based model trained specifically on legal documents in English. The model is trained from scratch on the 12GB of legal text in English, and follows the same configuration of BERT. Sentence-BERT (Reimers and Gurevych 2019) is another widespread general purpose model, that consists of a modification of BERT optimized for sentence similarity. We use 256 as the maximum token length and an embedding size of 384. For all models, we keep all default configuration hyper-parameters.

5.2 Data analysis

To better understand the complexity of the task and the distribution of the encoded data, we plot such data in two dimensions. We use PCA (Pearson 1901) to reduce to 50 dimensions and then t-SNE (van der Maaten and Hinton 2008) to reduce to 2 dimensions. The result is visible in Fig. 2. We observe that the negative samples are rather scattered, while the positive ones are more clustered. In other words, positive samples seem to be generally similar to each other, while negative samples do not. To validate this observation, we measure the intra- and inter-class similarity between the paragraphs. For each pair of paragraphs in the dataset, we measure the average cosine similarity between their embeddings.

The results, shown in Table 2, are consistent between the different splits and representations. Paragraphs belonging to the positive class are, on average, very similar to each other, contrary to negative samples. Indeed, both the intra-class similarity for the negative class and the inter-class similarity have low average values and high standard deviation. This means that, on average, the distance between a negative

[4] bert-base-uncased.

[5] nlpaueb/legal-bert-base-uncased.

[6] all-MiniLM-L6-v2.

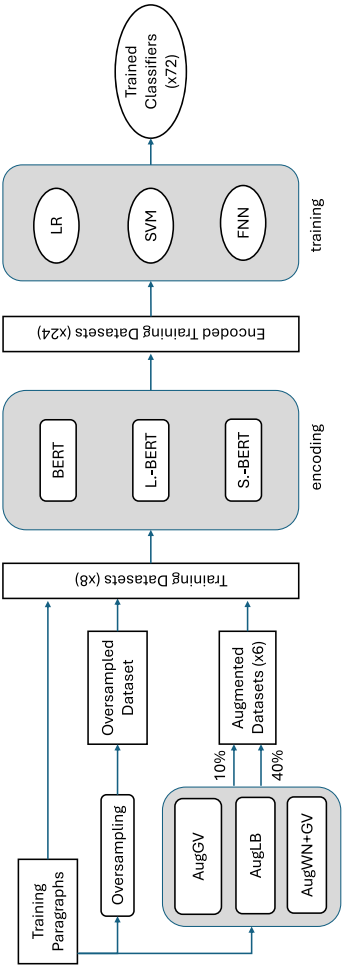


Fig. 1 Summary of the method used to train classifiers

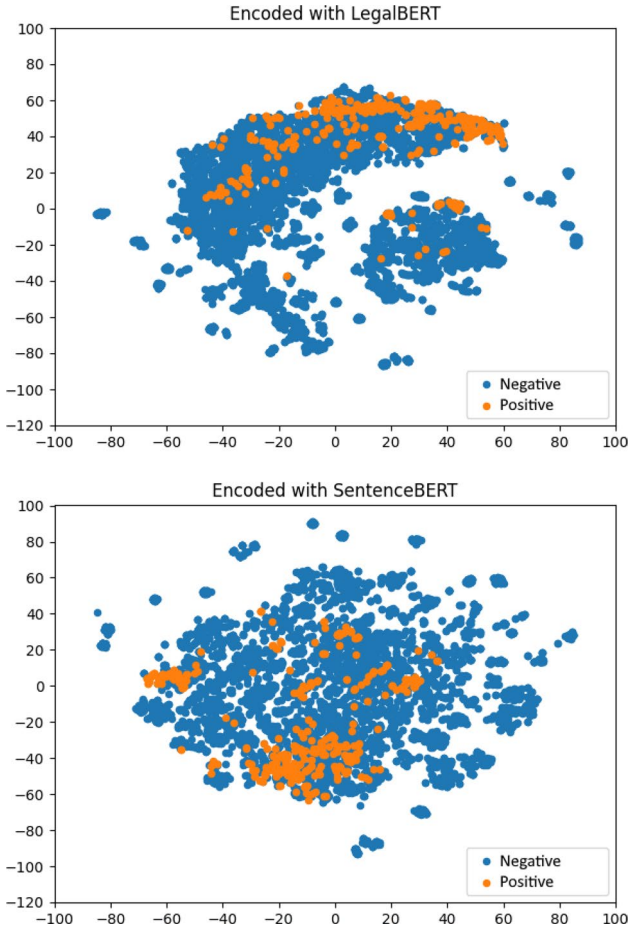


Fig. 2 Bi-dimensional representation of the samples in the training set

Table 2 Intra- and inter-class similarities between paragraphs. For each split and encoding are reported the average cosine similarity and the standard deviation

Encoding	Split	Intra Negative	Intra Positive	Inter-Class
Sentence-BERT	Train	0.30±0.15	0.43±0.14	0.33±0.14
Sentence-BERT	Val	0.29±0.15	0.42±0.14	0.31±0.14
Sentence-BERT	Test	0.28±0.15	0.44±0.12	0.32±0.14
BERT	Train	0.76±0.10	0.87±0.04	0.79±0.08
BERT	Val	0.75±0.10	0.86±0.04	0.78±0.09
BERT	Test	0.76±0.09	0.88±0.04	0.79±0.08
LegalBERT	Train	0.56±0.42	0.75±0.41	0.62±0.42
LegalBERT	Val	0.56±0.42	0.74±0.40	0.62±0.42
LegalBERT	Test	0.51±0.42	0.83±0.35	0.60±0.42

instance and a positive one is comparable with the distance between two negative instances.

This result aligns with our expectations. Indeed, paragraphs that contain legal principles tend to have a similar structure: they typically state that a certain legislative provision or judicial ruling should be interpreted in a certain way or that certain classifications should apply to relevant facts. Conversely, negative samples contain a variety of content, such as descriptions of facts of the case, hence it is harder to recognize patterns therein.

5.3 Class balancing and data augmentation

To balance the uneven distribution of data among the two classes and counter potential overfitting, we create several training datasets, experimenting with data oversampling and data augmentation. We oversample each training positive instance between 12 to 13 times, to obtain an equal number of positive and negative samples.

For data augmentation, we employ the three best strategies from Perçin et al. (2022) to create about 5,000 new synthetic samples out of the original positive instances. This amounts to replacing a set of terms in the original paragraphs with new terms that are semantically related. In our approach, the terms that can be selected for replacement are nouns, verbs, adverbs, and adjectives. We experiment with three replacement strategies and two different degrees of augmentation, replacing either 10% or 40% of terms of the paragraph.

- In the **AugGV** strategy the word is replaced with another word whose GloVe embedding (Pennington et al. 2014) is most similar.[7]
- The **AugLB** strategy relies on the LegalBERT (Chalkidis et al. 2020). The sample is given as input to the model, replacing the chosen word with the `MASK` token. The model then generate the replacement in place of the mask token.
- **AugWN+GV** is the most complex strategy, combining the use of the lexical database WordNet (Fellbaum 2010) with GloVe embeddings: we limit the replacement candidates to the WordNet synsets with a meaningful relationship and the related lemmas, then we choose the replacement according to the similarity of the embeddings.

This methodology results in the creation of 8 different training sets: the original one, the one obtained through oversampling, and 6 obtained using data augmentation. An example of data-augmented paragraph is reported in Fig. 3.

5.4 Classifiers

As classifiers, we use Support Vector Machine (SVM), Logistic Regression (LR), and Feed-Forward Neural Networks (FNN), as they are well-known and widely used classification methods with a low computational cost (Russell and Norvig 2010). For SVM, we set $C = 15$, while we keep all the remaining hyper-parameters as default.

[7] Similarity between embeddings is measured as cosine similarity.

Accordingly, although a measure by which the **office** authorities grant certain undertakings **favorable** tax treatment, which, although not involving the transfer of state resources, places the recipients in a more favourable financial **right** than other taxpayers, **quantities** to state aid within the meaning of article 107(1) tfeu, that is not the position as regards advantages resulting from a general measure applicable without distinction to all economic operators, which do not constitute state aid within the meaning of that provision (judgment in p, eu:c:2013:525, paragraph 18).

Accordingly, although a measure by which the **public** authorities grant certain undertakings **favourable** tax treatment, which, although not involving the transfer of state resources, places the recipients in a more favourable financial **position** than other taxpayers, **amounts** to state aid within the meaning of article 107(1) tfeu, that is not the position as regards advantages resulting from a general measure applicable without distinction to all economic operators, which do not constitute state aid within the meaning of that provision (judgment in p, eu:c:2013:525, paragraph 18).

Fig. 3 Example of a data-augmented paragraph (top) and its original version (bottom). Changed words are highlighted in bold. The augmented paragraph is generated with the AugGV 10% strategy. The original paragraph is extracted from the file "R2021_Commission v Poland.txt"

For LR, we set the number of iterations $max_iter = 500$ and keep the remaining default hyper-parameters. Lastly, we define the FNN as a three-layer dense network with hidden units equal to 128, 32, and 2, respectively. Each dense network has ReLU activation function. For each type of classifier, we train a model on each combination of the 3 different representations and 8 different training datasets. In total, we train 72 models, 24 for each type.

As baselines, we include two random classifiers with uniform and weighted probability, three classifiers trained on tf-idf encoding, and six classifiers based on cosine similarity and a threshold value. For the similarity baselines, a paragraph is classified as positive if its average similarity to all the positive samples of the training set is above a specific threshold.[8] Our previous analysis suggests that the similarity baseline may be able to retrieve most of the positive instances since such instances seem to be clustered, but would have difficulties in excluding the negative sample. We thus expect high recall and low precision for the positive class.

We also evaluated the zero-shot capability of state of the art LLMs to address the task. We considered three LLMs: Mistralv3, Llama3, and Phi3-mini.[9] We have chosen to use open-weight LLMs for the sake of reproducibility and given the cost of using proprietary solutions. Like with the classifiers developed by us, also LLMs were instructed to perform the legal principle extraction task by the paragraph. This ensured a fair comparison not only between LLMs and other methods and baselines, but also between LLMs, since the size of the input window required to carry out the task was suitable for all models and all inputs.

We used the following prompt:

[8] As threshold value we consider the average cosine similarity measured between the positive instances of the train set and the average minus the standard deviation.

[9] We used the HuggingFace implementation, specifically models mistralai/Mistral-7B-Instruct-v0.3, meta-llama/Meta-Llama-3.1-8B-Instruct, microsoft/Phi-3-mini-4k-instruct. We set the temperature to zero to exclude variations in the generated response since we prompt LLMs for a classification task. LLMs undergo a 4-bit quantization process to reduce computational overhead and fit single-GPU inference.

```
You are given a paragraph from a legal document representing a judicial decision on fiscal state aid by the Court of Justice of the European Union. Your task is to assess if the paragraph belongs to a legal principle as described below: A legal principle is a criterion that specifies what facts fall under a European law concept or determines the meaning of specific or general norms, as well as restricts or extends the application of such norms. Respond only YES or NO.
PARAGRAPH: { paragraph} ANSWER:
```

6 Experimental results

To determine the best approach for extracting legal principles, we experiment with every possible combination of representation, classifier, and data augmentation technique. We train the models by adopting early stopping (with 5 epochs of patience) on the objective loss measured on the validation set. We evaluate our models using the F1 score, the precision, and the recall metrics for the positive class.

Baselines and LLMs As shown in Table 3, the random baselines perform poorly. The baselines based on tf-idf have high precision but low recall, with the FNN obtaining the best F1 score of 0.54. The similarity-based baselines obtain higher recall and lower precision, as expected. Among them, the similarity baseline based on the average threshold and the BERT representation obtains the best F1 score, 0.50. As can be seen in Table 4, LLMs with zero-shot prompting do not perform better than the best baseline, with Llama obtaining the best result of 0.34. Given the high recall and low precision, such a score is mostly due to a high number of false positives.

Table 3 Classification performances of the baselines on the test set. The best F1 score is highlighted in bold

Method	F1	Precision	Recall
Random			
Uniform Probability	0.10	0.06	0.51
Weighted Probability	0.03	0.03	0.03
Tf-idf encoding			
LR	0.45	0.94	0.29
SVM	0.47	0.69	0.35
FNN	**0.54**	0.86	0.39
Similarity: AVG			
BERT	0.50	0.41	0.64
LEGAL-BERT	0.18	0.10	0.91
Sentence-BERT	0.37	0.28	0.53
Similarity: AVG-STD			
BERT	0.25	0.14	0.97
LEGAL-BERT	0.14	0.07	0.92
Sentence-BERT	0.16	0.09	1.00

Table 4 Classification performances of the LLM zero-shot approaches on the test set. The best F1 score is highlighted in bold

Method	F1	Precision	Recall
LLMs zero-shot			
Mistralv3	0.18	0.10	1.00
Llama3	**0.34**	0.22	0.73
Phi3-mini	0.21	0.12	0.97

Impact of oversampling and data augmentation For each combination of representation and classifier, we report in Table 5 its performance in the base case, with oversampling, and with one data augmentation method. Among the 6 possibilities presented in Section 5.3, we report the data augmentation method that obtained the best performance on the validation set. In most cases, SVM is the best-performing method and LR is the worst. As for the representations, BERT seems to outperform LEGAL-BERT, while they both outperform Sentence-BERT. In particular, LEGAL-BERT generally achieves higher recall than BERT but obtains a lower precision.

This result is coherent with our data analysis, where we observed a greater intra-class similarity between negative instances for BERT than LegalBERT. We hypothesize that this may be related to the fact that positive samples are not characterized

Table 5 Overview of the classification performances using machine learning approaches. We report the results on the test set of the best-performing models according to the validation set.The best F1, precision, and recall scores are highlighted in bold

Method	F1	Precision	Recall
BERT LR	0.63	0.69	0.58
+ oversampling	0.64	0.56	0.75
+ augmentation	0.64	0.54	0.78
BERT SVM	**0.68**	**0.79**	0.59
+ oversampling	0.64	0.78	0.54
+ augmentation	0.70	0.74	0.66
BERT FNN	0.64	0.66	0.62
+ oversampling	0.67	0.58	0.80
+ augmentation	0.66	0.56	0.79
LEGAL-BERT LR	0.63	0.70	0.57
+ oversampling	0.62	0.47	0.90
+ augmentation	0.61	0.47	0.89
LEGAL-BERT SVM	0.66	0.70	0.63
+ oversampling	0.66	0.56	0.81
+ augmentation	0.66	0.55	0.85
LEGAL-BERT FNN	0.66	0.66	0.66
+ oversampling	0.63	0.50	0.87
+ augmentation	0.62	0.48	0.89
Sentence-BERT LR	0.27	0.73	0.17
+ oversampling	0.52	0.36	**0.95**
+ augmentation	0.52	0.40	0.72
Sentence-BERT SVM	0.63	**0.79**	0.52
+ oversampling	0.63	**0.79**	0.52
+ augmentation	0.66	0.73	0.60
Sentence-BERT FNN	0.50	0.73	0.38
+ oversampling	0.57	0.44	0.83
+ augmentation	0.55	0.42	0.80

by domain-specific terminology but by a particular style or conceptual framing, as discussed in the guidelines. Therefore, a general-purpose model like BERT can generalize better than a domain-specific model. An additional element to consider is the pre-training set size of the two models, which is bigger for BERT. In most cases, oversampling and data augmentation are beneficial, increasing the recall and decreasing precision with respect to the base case, as it is reasonable to expect. The best combination overall appears to be BERT, SVM, and data augmentation.

Best performing methods To compare different data augmentation methods and determine the best approach for this dataset and task, we report in Table 6 the 10 best combinations of methods as measured on the test set. The use of BERT encoding and SVM classifiers clearly yields the best results, outperforming the alternatives by 5 points of F1 score. In particular, these methods are characterized by a higher precision (>0.70) rather than recall (<0.70). Conversely, using Legal-BERT leads to a higher recall and lower precision, as previously observed. Data augmentation contributes positively, with the best-performing method achieving an F1 score of 0.04 points better than the base case. For what concerns different augmentation techniques, the best result is obtained using AugGV and the lowest degree of augmentation, but we do not observe a consistent pattern between the three methods.

7 Extractive creation of collection of principles

The evaluation of binary classifications performed by our models enables us to determine in what cases each model has recognized or failed to recognize a principle in a paragraph, ignoring the magnitude of errors. Another interesting analysis focuses on the quality of the collection of paragraphs returned for a given decision. In fact, the system may succeed in producing a collection of paragraphs that, as a whole, contains all the relevant principles, even though the paragraphs in the collection may be different from those in the gold reference, since the same principle may be stated, maybe slightly differently, elsewhere in the decision. If we adopt a binary classification metric, as we did earlier, we would have a false positive and a false nega-

Table 6 Best 10 results obtained on the test set with any method. The Δ column reports the difference with the case that used the same configuration but no data augmentation

Encoding	Classifier	Augmentation	Aug %	F1	Δ F1	Precision	Recall
BERT	SVM	AugGV	10	**0.72**	**+0.039**	**0.80**	0.65
BERT	SVM	AugLB	10	0.70	+0.025	0.74	0.66
BERT	SVM	AugWN+GV	40	0.70	+0.020	0.80	0.62
BERT	SVM	AugWN+GV	10	0.69	+0.010	0.77	0.62
BERT	SVM	AugGV	40	0.68	+0.007	0.79	0.60
BERT	SVM	No	–	0.68	–	0.79	0.59
Legal-BERT	SVM	AugLB	10	0.67	+0.010	0.54	**0.88**
Legal-BERT	SVM	AugGV	40	0.67	+0.006	0.63	0.72
BERT	FNN	Oversampling	–	0.67	+0.030	0.58	0.80
Legal-BERT	SVM	AugLB	40	0.67	+0.005	0.55	0.85

tive, whereas a human judge may even fail to tell the original expert-made collection from the system's output. We then consider *collections* of automatically extracted paragraphs containing legal principles and compare them with the original collection of principles. We compare them in two ways: by ROUGE score, and by human evaluation.

7.1 Automatic evaluation

As a first evaluation, we compare the collection generated by our method (which thus contains both true positives and false positives) against the text composed only of the correct positive paragraphs. For both texts, the paragraphs are kept in the same order as in the original document.

We measure the ROUGE scores (Lin 2004) obtained for each test document as well as their average and report it in Table 7. As a comparison, we also include the average result obtained by Llama3 and the average score obtained by a baseline that randomly extracts a fixed number of paragraphs in the documents.[10]

The results are satisfactory, for all three metrics we obtain a F1 of 0.67 or higher, while the score of Llama and the baseline are inferior to, respectively, 0.45 and 0.10. The median values are between 0.74 and 0.82, indicating that most of the predictions are of very high quality, except for a couple of outliers.

7.2 Human evaluation

ROUGE only measures lexical overlap, and to the best of our knowledge, there is no automatic method for the evaluation of legal adequacy that can be considered robust and reliable. Consequently, we also perform a human evaluation. Since humans may have a prejudice regarding automatically generated content (Harasta et al. 2024), instead of simply asking humans to rate the automatically generated collections, We ask 2 human experts to distinguish them from the original collection of principles. In this way, the evaluation is not affected by human biases. We have chosen the experts so that their profile is similar to the annotators' one: they hold a PhD in Law and have an expertise in this domain. For the sake of reproducibility, something too often overlooked in NLP research (Belz et al. 2023), all the material used for the human evaluation is included in our repository.

7.2.1 Evaluation setting

We task our expert evaluators to discern the automatically generated collections from the collections made by other human experts. We interpret a low human performance on this task as an indication that our method has created a plausible output. We structure the human evaluation in two stages. In the first stage, the experts are given a set of documents, which may be real or automatically generated. The experts have to tell which one. The purpose here is to evaluate whether our automatically generated collections of principles would be considered plausible by a human expert. In the

[10] Such a number is chosen as the median number of positive samples per document in the training set.

Table 7 Evaluation of the extractive creation of collections of principles on the test set

	ROUGE-1			ROUGE-2			ROUGE-L		
DOC ID	F1	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec
00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
01	0.87	0.80	0.95	0.83	0.77	0.91	0.82	0.76	0.90
02	0.46	1.00	0.30	0.45	1.00	0.29	0.46	1.00	0.30
03	0.74	0.83	0.67	0.59	0.66	0.54	0.52	0.58	0.47
04	0.60	0.47	0.81	0.44	0.35	0.60	0.35	0.28	0.48
05	0.64	1.00	0.47	0.64	1.00	0.47	0.64	1.00	0.47
06	0.84	1.00	0.73	0.84	1.00	0.73	0.84	1.00	0.73
07	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
08	0.80	0.95	0.70	0.74	0.88	0.64	0.70	0.83	0.61
09	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
10	0.82	1.00	0.69	0.82	1.00	0.69	0.82	1.00	0.69
11	0.90	0.86	0.96	0.87	0.82	0.92	0.86	0.82	0.91
12	0.86	0.93	0.79	0.78	0.85	0.72	0.74	0.81	0.69
AVERAGE									
Our Method	0.73	0.83	0.70	0.69	0.79	0.66	0.67	0.77	0.63
Baseline	0.08	0.29	0.05	0.07	0.28	0.05	0.07	0.28	0.05
LLM Llama3	0.42	0.30	0.95	0.38	0.26	0.86	0.35	0.24	0.80

second stage, the experts are provided with both the automatically generated and the original versions of the documents, without knowing which is which, and must tell them apart. In this setting the experts can exploit the additional knowledge given by the direct comparison of the two documents. In either stage, the two experts operate independently of each other, and are not told the source of the documents.

We run this evaluation task on the entire validation and test sets, for a total of 26 documents out of 64. We intentionally excluded the training set documents as standard practice to test the ability of the model to generalize. We use the F1 score on the "automatically generated" class as the evaluation metric.

7.2.2 Results and discussion

During the first stage, on average, experts obtained an F1 score of 0.40, with higher precision (0.56) and lower recall (0.31), indicating that their errors are mostly misclassification of automatically generated collections of principles as original ones. During the second stage, they changed their opinion on a few cases, and correctly identified a higher number of documents, obtaining average scores of 0.66 for F1, 0.73 for precision, and 0.62 for recall.[11] These results show that the documents created through our best model are plausible enough to pass expert scrutiny. They also demonstrate that distinguishing them from the original ones is non-trivial.

[11] In the few cases where the two principles were identical, they were marked both as original.

7.2.3 Error analysis

Finally, we revealed to legal experts which version had been generated using our method and asked them to comment on the differences between the collections. With regard to false positives, the extracted collections occasionally extend beyond the interpretation of core legal concepts. Instead, they include discussions of appeal procedures, partial restatements of facts or descriptive rather than interpretative references to case law (e.g., C-390/99). In other instances, false positives appear in paragraphs primarily describing the factual background rather than legal interpretation (e.g., Joined Cases C-164/15 P and C-165/15 P). Several false positives contain phrases and expressions also present in principles-related passages within the same judgment, requiring close contextual reading to distinguish between them. One possible explanation is the recurrence of similar terminology across both relevant and irrelevant paragraphs. False negatives, by contrast, occasionally involve passages in which factual discussion is intertwined with the interpretation of the legal principle, potentially reducing detectability (e.g., C-83/09 P). In some cases, discussions of principles grounded in earlier case law appear overly abstract or insufficiently explicit for reliable identification. Sometimes, the same judgment contains both false positives and false negatives in similar or adjacent paragraphs (e.g., C-333/07, C-449/14). This may suggest a limitation in our paragraph-wise classification approach, whereas an expert can make a more informed decision by considering a larger context. Overall, the extraction of complex reasoning passages such as legal principles represents a demanding benchmarking task, as the indicators associated with relevant and irrelevant paragraphs may overlap due to the highly formalised nature of legal language.

8 Limitations, impact, and ethics

These methods are aimed at supporting legal practitioners, rather than substituting them. We are aware that there is always the risk of users excessively relying on a fallible system. Users should be provided appropriate guidance on the limitations of the system and the need to check its outcomes.

We believe that our methods can have a positive ethical-legal impact by inducing practitioners to consider a larger set of principles in addressing concrete cases, which may contribute to more informed decisions, taking into account a larger set of ethically and legally relevant aspects. Our methods can also contribute to the legal process by showing the dynamical development of the case law, and thus directing practitioners to prioritize more recent principles and interpretations.

Biases and representativeness We are aware that annotation can be biased by the interests of the annotators, but we believe that the identification of principles, as we have understood them, can be in general objective, in particular since (i) we marked all principles rather than only focusing on the most controversial or innovative ones, (ii) annotators were given clear and univocal guidelines for identifying principles. We are also aware that Fiscal State aid decisions are a small and specialized subset of

the CJEU case law and, therefore, might not be representative of the court's reasoning in general.

LLMs limitations In our comparison with LLM performance, we crafted a prompt by following a standard methodology and adopted zero-shot prompting. The intended meaning of such a comparison is to estimate what results could be obtained with simple zero-shot prompts, not to investigate the limits of LLMs or the impact of other prompting techniques, such as few-shot prompting and chain-of-thought. We leave such an investigation as future work, as it deserves a separate study. Indeed, making a fair comparison between LLMs and classification models like ours is non-trivial due to their being highly dependent on the formulation of the prompt and probabilistic nature. Slight variations in the prompt may lead to significantly different results (Gan and Mori 2023; Lu et al. 2022; Martínez 2024; Salinas and Morstatter 2024; Sclar et al. 2024). Moreover, recent studies have advised against the incautious use of LLM in the legal domain due to their extensive tendency to hallucinate (Dahl et al. 2024), hence the focus of our work on methods other than LLM prompting.

9 Conclusion

We have provided some novel results in the automated analysis of case law: a methodology for identifying and extracting legal principles (also applicable to civil law jurisdictions), an evaluation of using data augmentation for this task, a comparison of the performance of different tools. For this purpose, we constructed a novel annotated corpus based on documents of the Court of Justice of the European Union. Our experiments on sentence classification show that data augmentation is beneficial and that BERT encoding together with an SVM classifier leads to the best result, obtaining a F1 score of 0.72. Moreover, we show that such a classifier is able to extract and aggregate paragraphs to create meaningful collections of legal principles.

In the future, we plan to refine our extraction technique to address not only paragraphs but also single sentences or spans of text. We also plan to expand our analysis to other languages, so as to test the scalability of our method, as experimented in other studies (Galassi et al. 2025). In this regard, the adoption of standard multilingual representation methods may be sufficient for the task at hand since the CJEU decisions are available in all EU languages; adopt a consistent drafting style across them (e.g., keeping the same paragraph numberings), and use multilingual legal dictionaries for many legal concepts. Moreover, the transfer of legal principle extraction and analysis from one area of law to another or from one jurisdiction to another is still an open research challenge, that we would like to investigate in our future work. This would make it possible to compare the content, form, role, and evolution of principles in different areas of the law.

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- Generative AI for Law (CP FIS-2024-07323, CUP J53C25002490001); POLINE - Principles Of Law In National and European VAT (European Union's Justice Programme, G.A. No.: 101087342); PRIN 2017 LAILA - Legal Analytics for Italian Law (Ref. Prot n. 20174F958N); the Jean Monnet Network Project European Network on Digitalization and E-Governance (ENDE), Grant Agreement No. 101127038 (EACEA, Erasmus+).

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Data and Code Availability All materials utilized in this work, including code and data, are publicly available at https://github.com/nlp-unibo/legal-principles-cjeu.

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References

Abonizio HQ, Junior SB (2020) Pre-trained data augmentation for text classification. In: BRACIS, Lecture notes in computer science, vol 12319. Springer, pp 551–565. https://doi.org/10.1007/978-3-030-61377-8_38

Achena F, Preti D, Venditti D et al (2023) Legal summarization: to each court its own model. In: CLiC-it. https://aclanthology.org/2023.clicit-1.3/

Alexy R (2000) On the structure of legal principles. Ratio Jurix 13:294–304

Anaby-Tavor A, Carmeli B, Goldbraich E et al (2020) Do not have enough data? deep learning to the rescue! Proceed AAAI Conf Artif Intell 34:7383–7390. https://doi.org/10.1609/aaai.v34i05.6233

Belz A, Thomson C, Reiter E et al (2023) Non-repeatable experiments and non-reproducible results: The reproducibility crisis in human evaluation in NLP. In: Rogers A, Boyd-Graber J, Okazaki N (eds) Findings of the Association for Computational Linguistics: ACL 2023. Association for Computational Linguistics, Toronto, Canada, pp 3676–3687. https://doi.org/10.18653/v1/2023.findings-acl.226, https://aclanthology.org/2023.findings-acl.226

Branting L (1993) A computational model of ratio decidendi. Artif Intell Law 2(1):1–33

Branting K (1994) Four challenges for a computational model of legal precedent. THINK (J Inst Lang Technol Artif Intell) 3:62–69

Chalkidis I, Fergadiotis M, Malakasiotis P et al (2020) LEGAL-BERT: The muppets straight out of law school. In: EMNLP (Findings). Association for Computational Linguistics, Online, pp 2898–2904

Chalkidis I, Jana A, Hartung D et al (2022) Lexglue: A benchmark dataset for legal language understanding in english. In: ACL. Association for Computational Linguistics, pp 4310–4330

Cheung TH, Yeung DY (2021) MODALS: Modality-agnostic automated data augmentation in the latent space. In: International conference on learning representations. https://openreview.net/forum?id=XjYgR6gbCEc

Cohen J (1960) A coefficient of agreement for nominal scales. Educ Psychol Measur 20:37–46. https://doi.org/10.1177/001316446002000104

Dahl M, Magesh V, Suzgun M et al (2024) Large legal fictions: Profiling legal hallucinations in large language models. J Legal Anal 16(1):64–93. https://doi.org/10.1093/jla/laae003. https://academic.oup.com/jla/article-pdf/16/1/64/58336922/laae003.pdf
Devlin J, Chang M, Lee K, et al (2019) BERT: pre-training of deep bidirectional transformers for language understanding. In: NAACL-HLT. Association for Computational Linguistics, pp 4171–4186
Duxbury N (2008) The Nature and the Authority of Precedent. Cambridge University Press
Dworkin RM (1977) Taking Rights Seriously. Duckworth
Fellbaum C (2010) WordNet. Springer, Netherlands, Dordrecht, pp 231–243
Feng SY, Gangal V, Wei J et al (2021) A survey of data augmentation approaches for NLP. https://doi.org/10.18653/v1/2021.findings-acl.84, https://aclanthology.org/2021.findings-acl.84
Galassi A, Lippi M, Torroni P (2021) Attention in natural language processing. IEEE Trans Neural Networks Learn Syst 32(10):4291–4308. https://doi.org/10.1109/TNNLS.2020.3019893
Galassi A, Lagioia F, Jablonowska A et al (2025) Unfair clause detection in terms of service across multiple languages. Artif Intell Law 33(3):641–689. https://doi.org/10.1007/S10506-024-09398-7
Gan C, Mori T (2023) Sensitivity and robustness of large language models to prompt template in japanese text classification tasks. In: PACLIC. Association for computational linguistics, pp 1–11
Ghosh S, Evuru CKR, Kumar S et al (2023) DALE: Generative data augmentation for low-resource legal NLP. In: Bouamor H, Pino J, Bali K (eds) Proceedings of the 2023 conference on empirical methods in natural language processing. Association for Computational Linguistics, Singapore, pp 8511–8565. https://doi.org/10.18653/v1/2023.emnlp-main.528, https://aclanthology.org/2023.emnlp-main.528
Goldberg Y (2018) Neural network methods for natural language processing, vol 44. MIT Press
Goodhart AL (1959) Ratio decidendi of a case. Mod Law Rev 21:117–24
Grabmair M, Ashley KD, Chen R et al (2015) Introducing LUIMA: an experiment in legal conceptual retrieval of vaccine injury decisions using a UIMA type system and tools. In: Proceedings of the 15th international conference on artificial intelligence and law, pp 69–78
Gray M, Savelka J, Oliver W et al (2022) Toward automatically identifying legally relevant factors. In: JURIX. IOS Press, pp 53–62. https://doi.org/10.3233/FAIA220448
Gray MA, Savelka J, Oliver WM et al (2024) Empirical legal analysis simplified: reducing complexity through automatic identification and evaluation of legally relevant factors. Phil Trans R Soc A 382(2270):20230155
Grundler G, Santin P, Galassi A et al (2022) Detecting arguments in CJEU decisions on fiscal state aid. In: Proceedings of the 9th workshop on argument mining, ArgMining@COLING. International conference on computational linguistics, pp 143–157. https://aclanthology.org/2022.argmining-1.14
Guo Z, Liu J, He T et al (2020) Taujud: test augmentation of machine learning in judicial documents. In: ISSTA. ACM, pp 549–552. https://doi.org/10.1145/3395363.3404364
Gupta R (2019) Data augmentation for low resource sentiment analysis using generative adversarial networks. In: ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp 7380–7384. https://doi.org/10.1109/ICASSP.2019.8682544
Habernal I, Faber D, Recchia N et al (2023) Mining legal arguments in court decisions. Artif Intell Law. https://doi.org/10.1007/s10506-023-09361-y
Harasta J, Novotná T, Savelka J (2024) It cannot be right if it was written by ai: on lawyers' preferences of documents perceived as authored by an llm vs a human. Artif Intell Law 1–38. https://doi.org/10.1007/s10506-024-09422-w
Hu EJ, Shen Y, Wallis P et al (2022) Lora: Low-rank adaptation of large language models. ICLR 1(2):3
Jin D, Jin Z, Zhou JT et al (2020) Is BERT really robust? A strong baseline for natural language attack on text classification and entailment. https://doi.org/10.1609/AAAI.V34I05.6311
Jurafsky D, Martin JH (2025) Speech and language processing: An introduction to natural language processing, computational linguistics, and speech recognition with language models, 3rd edn. https://web.stanford.edu/~jurafsky/slp3/, online manuscript released August 24, 2025
Kumar V, Choudhary A, Cho E (2021) Data augmentation using pre-trained transformer models. arXiv:2003.02245
Lamond G (2005) Do precedents create rules. Legal Theory 1–26
Lin CY (2004) ROUGE: A package for automatic evaluation of summaries. Text Summarization Branches Out. Association for Computational Linguistics, Barcelona, Spain, pp 74–81
Lippi M, Torroni P (2016) Argumentation mining: State of the art and emerging trends. ACM Trans Internet Technol 16(2). https://doi.org/10.1145/2850417
Liu P, Yuan W, Fu J et al (2023) Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing. ACM Comput Surv 55(9):1–35

Lu Y, Bartolo M, Moore A et al (2022) Fantastically ordered prompts and where to find them: overcoming few-shot prompt order sensitivity. In: ACL (1). Association for computational linguistics, pp 8086–8098. https://doi.org/10.18653/v1/2022.acl-long.556

Lupu Y, Voeten E (2012) Precedent in international courts: A network analysis of case citations by the european court of human rights. British J Politic Sci 42(2):413–439

MacCormick DN (2005) Using precedents. In: Rhetoric and the rule of law. Oxford University Press

Martínez E (2024) Re-evaluating gpt-4's bar exam performance. Artif Intell Law. https://doi.org/10.1007/s10506-024-09396-9

Medvedeva M, Vols M, Wieling M (2020) Using machine learning to predict decisions of the european court of human rights. Artif Intell Law 28:237–266. https://doi.org/10.1007/s10506-019-09255-y

Min J, McCoy RT, Das D, et al (2020) Syntactic data augmentation increases robustness to inference heuristics. In: Jurafsky D, Chai J, Schluter N et al (eds) Proceedings of the 58th annual meeting of the association for computational linguistics. Association for computational linguistics, Online, pp 2339–2352. https://doi.org/10.18653/v1/2020.acl-main.212, https://aclanthology.org/2020.acl-main.212

Mochales R, Moens MF (2011) Argumentation mining. Artif Intell Law 19:1–22

Mumford J, Atkinson K, Bench-Capon T (2022) Reasoning with legal cases: a hybrid adf-ml approach. In: JURIX. IOS Press, pp 93–102. https://doi.org/10.3233/FAIA220452

Pearson K (1901) On lines and planes of closest fit to systems of points in space. London Edinburgh Dublin Philosoph Magaz J Sci 2(11):559–572

Pennington J, Socher R, Manning CD (2014) Glove: Global vectors for word representation. In: EMNLP, pp 1532–1543

Perçin S, Galassi A, Lagioia F et al (2022) Combining wordnet and word embeddings in data augmentation for legal texts. In: Proceedings of the natural legal language processing workshop, NLLP@EMNLP. Association for Computational Linguistics, pp 47–52. https://doi.org/10.18653/V1/2022.NLLP-1.4

Peric L, Mijic S, Stammbach D et al (2020) Legal language modeling with transformers. In: Ashley KD, Atkinson K, Branting LK et al (eds) Proceedings of the fourth workshop on automated semantic analysis of information in legal text held online in conjunction with the 33rd international conference on legal knowledge and information systems, ASAIL@JURIX 2020, December 9, 2020, CEUR Workshop Proceedings, vol 2764. CEUR-WS.org. http://ceur-ws.org/Vol-2764/paper2.pdf

Plug J (2000) Indicators of obiter dicta. a pragma-dialectical analysis of textual clues for the reconstruction of legal argumentation. Artif Intell Law 8(2):189–203. https://doi.org/10.1023/A:1008327715564

Poudyal P, Savelka J, Ieven A et al (2020) ECHR: legal corpus for argument mining. In: Cabrio E, Villata S (eds) Proceedings of the 7th workshop on Argument Mining, ArgMining 2020, Barcelona, Spain (Online), December 13, 2020. Association for Computational Linguistics, pp 67–75. https://aclanthology.org/2020.argmining-1.8/

Preti D, Giannone C, Favalli A et al (2024) Automatic summarization of legal texts, extractive summarization using llms. In: Proceedings of the Ital-IA 2024: 4th National conference on artificial intelligence. https://ceur-ws.org/Vol-3762/501.pdf

Raz J (1972) Legal principles and the limits of law. Yale Law J 81:823–54

Reimers N, Gurevych I (2019) Sentence-bert: sentence embeddings using siamese bert-networks. In: EMNLP-IJCNLP. Association for computational linguistics, pp 3980–3990

Russell SJ, Norvig P (2010) Artificial Intelligence - A Modern Approach. Third International Edition, Pearson Education

Šadl U, Olsen HP (2017) Can quantitative methods complement doctrinal legal studies? using citation network and corpus linguistic analysis to understand international courts. Leiden J Intern Law 30(2):327–349

Salinas A, Morstatter F (2024) The butterfly effect of altering prompts: how small changes and jailbreaks affect large language model performance. In: Findings of the association for computational linguistics ACL. Association for computational linguistics pp 4629–4651. https://doi.org/10.18653/v1/2024.findings-acl.275

Sansone C, Sperlí G (2022) Legal information retrieval systems: State-of-the-art and open issues. Inf Syst 106:101967

Santin P, Grundler G, Galassi A et al (2023) Argumentation structure prediction in CJEU decisions on fiscal state aid. In: ICAIL. ACM, pp 247–256. https://doi.org/10.1145/3594536.3595174

Schauer F (1991) Playing by the Rules. Clarendon

Schön W (2016) Tax legislation and the notion of fiscal aid: a review of 5 years of european jurisprudence. State aid law and business taxation pp 3–26

Sclar M, Choi Y, Tsvetkov Y et al (2024) Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. In: ICLR

Sheik R, Sundara K, Nirmala J (2024) Neural data augmentation for legal overruling task: Small deep learning models vs. large language models. Neural Process Lett 56. https://doi.org/10.1007/s11063-024-11574-4

Shorten C, Khoshgoftaar TM, Furht B (2021) Text data augmentation for deep learning. J Big Data 8(1):101

Shulayeva O, Siddharthan A, Wyner A (2017) Recognizing cited facts and principles in legal judgements. Artif Intell Law 25(1):107–126. https://doi.org/10.1007/s10506-017-9197-6

Valvoda J, Ray O (2017) From case law to ratio decidendi. JSAI-isAI Workshops. lecture notes in computer science, vol 10838. Springer, pp 20–34. https://doi.org/10.1007/978-3-319-93794-6_2

van der Maaten L, Hinton G (2008) Visualizing data using t-sne. J Mach Learn Res 9(86):2579–2605

Vaswani A, Shazeer N, Parmar N et al (2017) Attention is all you need. In: NeurIPS, pp 5998–6008

Wang WY, Yang D (2015) That's so annoying!!!: A lexical and frame-semantic embedding based data augmentation approach to automatic categorization of annoying behaviors using #petpeeve tweets. In: Màrquez L, Callison-Burch C, Su J (eds) Proceedings of the 2015 conference on empirical methods in natural language processing. Association for Computational Linguistics, Lisbon, Portugal, pp 2557–2563. https://doi.org/10.18653/v1/D15-1306, https://aclanthology.org/D15-1306

Wei J, Zou K (2019) EDA: Easy data augmentation techniques for boosting performance on text classification tasks. In: Inui K, Jiang J, Ng V, et al (eds) Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP). Association for Computational Linguistics, Hong Kong, China, pp 6382–6388. https://doi.org/10.18653/v1/D19-1670, https://aclanthology.org/D19-1670

Wu X, Lv S, Zang L et al (2019). Conditional BERT contextual augmentation. https://doi.org/10.1007/978-3-030-22747-0_7

Yan G, Li Y, Zhang S et al (2019) Data augmentation for deep learning of judgment documents, lecture notes in computer science, vol 11936, Springer, pp 232–242. https://doi.org/10.1007/978-3-030-36204-1_19

Zhou W, Zhang S, Gu Y et al (2024) Universalner: Targeted distillation from large language models for open named entity recognition. In: The twelfth international conference on learning representations, ICLR 2024, Vienna, Austria, May 7-11, 2024. OpenReview.net. https://openreview.net/forum?id=r65xfUb76p

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